

Strata Cadastre

Problem Statement ID	26011
Problem Statement Title	3D ULPIN Generation and Vertical Property Mapping System
Theme	Smart Automation
PS Category	Software
Team ID	—
Team Name	Strata_Cadastre_KP

PROPOSED SOLUTION

Giving every flat, basement, utility line and air-rights volume its own legally distinct 3D identity.

The gap

India's land records identify PARCELS OF GROUND. A 12-storey building is one parcel — 24 flats share a single 2D ULPIN. Basements, parking, air rights and utility runs have no unique identity at all.

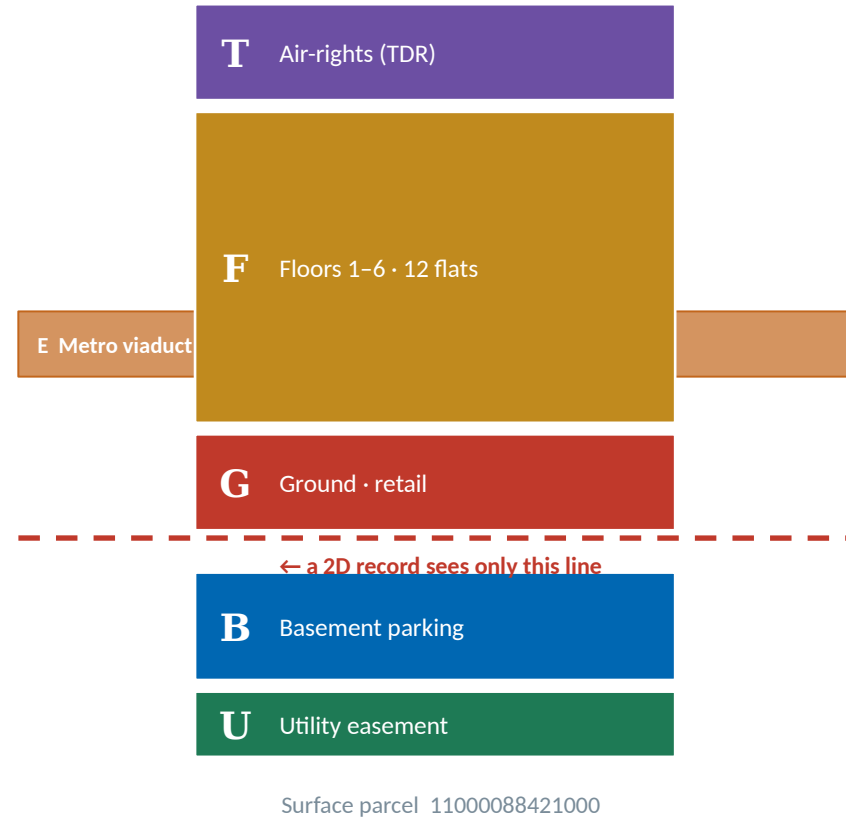
Why it matters

Overlapping claims cannot be DETECTED — only litigated after the fact. Undocumented subsurface utility runs are a leading cause of excavation strikes.

Our solution

A volumetric cadastre. Every unit is stored as a true closed 3D volume with its own 28-character 3D-ULPIN, and the database physically REFUSES to register a volume that overlaps an existing one.

ONE PARCEL, TEN REGISTERED VOLUMES



3D-ULPIN · 28 characters

240124	11000088421000	T	906	T010
LGD	2D ULPIN	TYPE	FLOOR	UNIT

WHAT CHANGES

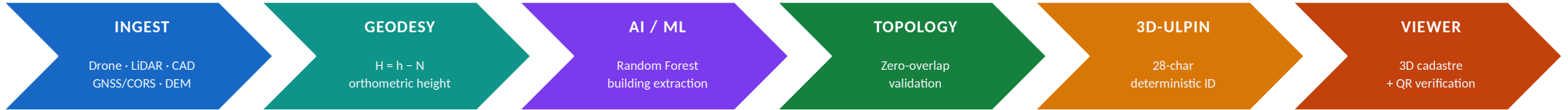
2D record today	Strata Cadastre
24 flats, one ULPIN	One 3D-ULPIN per unit
Parking undocumented	Bay registered as a volume
Air rights unrecordable	TDR volume with its own ID
Utility runs invisible	Easement volume, AMC-held
Overlaps found in court	Overlap rejected on entry
Deviation found on complaint	ΔV computed automatically

SIX VERTICAL PROPERTY TYPES

T Air-rights	E Elevated	F Floors
G Ground	B Basement	U Utility

TECHNICAL APPROACH

PROCESSING PIPELINE

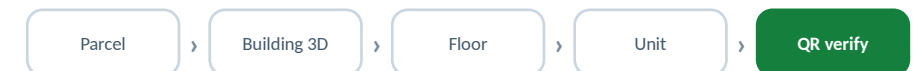


SYSTEM ARCHITECTURE — LAYERED STACK

FRONTEND	React + CesiumJS · Three.js / WebGL · QR verification
BACKEND	FastAPI · async REST · OGC API-Features aligned
DATABASE	PostgreSQL + PostGIS 3D + SFCGAL · PolyhedralSurfaceZ
ULPIN	28-character deterministic generator · 6 vertical types
AI / ML	scikit-learn Random Forest · 8 geometric features
PROCESSING	Python · NumPy · SciPy · OpenCV · PDAL / GDAL
INGESTION	LAS/LAZ · PLY · GeoTIFF · DXF · GeoJSON · ASCII Grid



USER JOURNEY


99.39%

ML accuracy

1.56%

deviation error

±1 cm

plinth

0.022°

ICP

6 / 6

vertical types

FEASIBILITY AND VIABILITY

ECONOMIC FEASIBILITY

■ No re-survey needed

Extends the existing 14-digit ULPIN, so adoption starts from records already digitised.

■ Broadens the tax base

Volumetric assessment on actual cubic metres and usage — without raising rates.

■ Recovers lost revenue

Automated deviation detection surfaces regularisation cases found today only by complaint.

TECHNICAL FEASIBILITY

■ Working prototype

Full pipeline runs end to end — ingest to ULPIN to topology to 3D viewer, with an automated test suite.

■ Mature open standards

ISO 19152 LADM, OGC CityGML 3.0, 3D Tiles, COPC/EPT. No unproven technology in the stack.

■ Commodity hardware

Self-hosted Docker stack runs inside NIC or state data centres. No external cloud needed.

LEGAL FEASIBILITY

■ Matches Indian apartment law

Models Undivided Share in Land plus exclusive right to a spatial volume as two linked records.

■ DPDP Act 2023 compliant

Citizen and Registrar tiers built into the data model — owner KYC and liens never reach the public tier.

■ Extends, never replaces

The statutory 2D ULPIN stays authoritative; 3D units hang off it as child records.

SOCIAL FEASIBILITY

■ Protects flat buyers

Verifiable spatial identity, RERA status and encumbrance check before purchase.

■ Cuts litigation at source

Overlapping claims become impossible to register rather than arguable in court.

■ Public safety

Registered subsurface easements prevent the excavation strikes undocumented utility runs cause.

KEY RISKS AND HOW WE OVERCOME THEM

Sub-metre LiDAR is scarce

Phone photogrammetry — 200 photos to Meshroom gives a real point cloud. No permit, no cost.

Legacy 2D records inaccurate

Precedence matrix plus per-node uncertainty. Legacy data is flagged, never overwritten.

SFCGAL not on every host

Verified path: Neon and Debian PostGIS images work. A 9-check suite proves any database.

Municipal adoption effort

Phased rollout on dense urban parcels first; rural 2D records stay untouched.

DETAILED FEASIBILITY REPORT

Validated: ULPIN generation · topology rejection · deviation detection (1.56%) · ML classifier (99.39%) · 9-check DB suite.
Not yet: ML trained on synthetic data, not field-labelled LiDAR — retraining is the next step.

IMPACT AND BENEFITS

10

ownership volumes recorded for one parcel (2D records: 1)

528 m³

unauthorised construction detected automatically

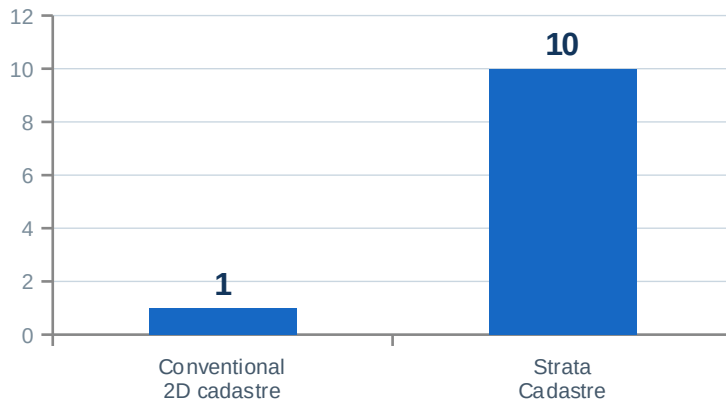
99.39%

ML classifier accuracy on held-out test data

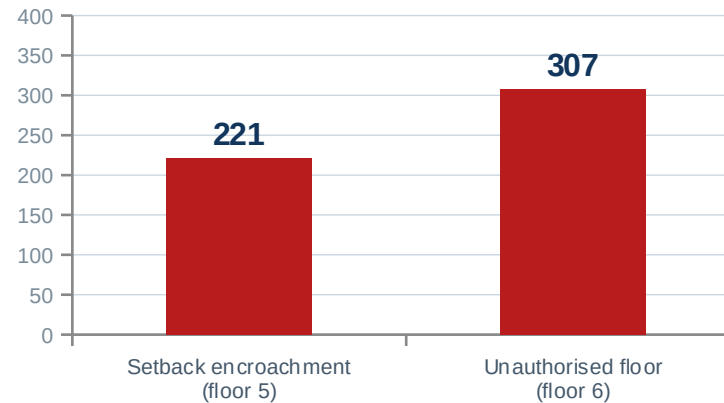
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vertical property types generated and validated

WHAT ONE PARCEL CAN LEGALLY HOLD



UNAUTHORISED VOLUME DETECTED (M³)



PROTOTYPE & FIELD EVIDENCE

Drop screenshots or field photos here

prototype view · site visit · sanctioned plan

IMPACT ON THE TARGET AUDIENCE

1

Flat owners & buyers

Verifiable spatial identity, RERA status and encumbrance check before purchase. Parking, basement and terrace rights become individually titled.

2

Registrar & Revenue offices

Overlapping claims become impossible to REGISTER rather than arguable in court. Mutation and lifecycle tracked per volume.

3

Municipal corporations

Volumetric tax on actual cubic metres and usage. Unauthorised construction detected automatically, not on complaint.

4

Utilities & infrastructure

Subsurface and airspace rights get registered volumes — preventing excavation strikes and alignment disputes.

SOCIAL

Cuts the root cause of apartment litigation; protects buyers before purchase

ECONOMIC

Volumetric tax base; recovered regularisation revenue; cleaner titles cut lending risk

GOVERNANCE

Registered subsurface and airspace rights enable genuine 3D urban planning

RESEARCH AND REFERENCES

STANDARDS & FRAMEWORKS

ISO 19152 — LADM

Land Administration Domain Model

iso.org/standard/51206.html

OGC CityGML 3.0

LOD2 / LOD3 volumetric city objects

ogc.org/standard/citygml

OGC 3D Tiles

Streaming large 3D geospatial content

ogc.org/standard/3dtiles

COPC

Cloud Optimised Point Cloud

copc.io

OGC API — Features

REST interface our API aligns to

ogcapi.ogc.org/features

GOVERNMENT & POLICY

DILRMP

Land Records Modernisation Programme

dilrmp.gov.in

Dept. of Land Resources

ULPIN — the 14-digit ID we extend

dolr.gov.in

SVAMITVA Scheme

Drone survey of inhabited areas

svamitva.nic.in

DPDP Act 2023

Basis for Citizen / Registrar tiers

meity.gov.in

AnyRoR Gujarat

State land record portal

anyror.gujarat.gov.in

TECHNOLOGY STACK

PostGIS + SFCGAL

3D volume operations

postgis.net/docs/reference.html

scikit-learn

Random Forest classifier

scikit-learn.org

PDAL / GDAL

Point cloud & raster processing

pdal.io · gdal.org

OpenCV

Floor plan extraction

opencv.org

CesiumJS / Three.js

3D web rendering

cesium.com · threejs.org

FastAPI

Async REST backend

fastapi.tiangolo.com

PRIMARY FIELD RESEARCH — AHMEDABAD

City Mamlatdar & Executive Magistrate — possession and easement dispute process under the Mamlatdar's Courts Act, 1906 · City Survey Office — Property Card record format and its missing vertical fields · Local developer — sanctioned plan structure, UDS allotment and BU-permission workflow